

FlowSentinel AI: Step-by-Step Setup & Dashboard Guide

Prepared by CTech Data Engineers for Demo Presenters & Cloud Admins

Do I Need a Dashboard to Monitor? (Dashboard Options)

YES! A Dashboard is Essential for Human-in-the-Loop 1-Click Governance

Dashboard Tool Option	Setup Effort	Best Use Case in Demo
1. Built-in Custom Web App (React/Vite + FastAPI) ⭐	Low (Included)	Interactive 1-Click Approval Deck for Oncall Engineers (approve PRs, run backfills).
2. GCP Looker Studio (100% Free)	Zero	Executive Compliance & SLA Scorecard connecting to BigQuery de_ops_metadata .
3. GitHub Projects v2 Kanban Board (100% Free)	Zero	Code-Native Ticket Tracking (Incident Triage → PR Drafted → Resolved).

Step-by-Step Demo Provisioning & Execution Instructions

Phase 1: Environment Configuration (5 Mins)

Set project environment variables in terminal:

```
export GCP_PROJECT_ID="ctech-flowsentinel-demo-dev"
export GITHUB_PAT_TOKEN="ghp_your_generated_token"
export GITHUB_REPO="ctech-flowsentinel-demo"
```

Phase 2: One-Click GCP Infrastructure Provisioning (5 Mins)

Run master provisioning script to enable APIs, BigQuery datasets, Pub/Sub, and Cloud Logging sinks:

```
chmod +x ./setup/gcp_provisioning.sh && ./setup/gcp_provisioning.sh
```

Phase 3: Launching Demo Suite & Dashboard

Execute master interactive demo launcher suite:

```
./run_all_demos.sh A
```

Demo Module	Manual Command	What It Demonstrates
Demo 1: Autonomous Triage	cd demos/demo_1_autonomous_triage && ./run_demo.sh	Schema drift failure detection → Gemini 2.0 RCA → GitHub Issue #42.
Demo 2: Code Patch & Approval	cd demos/demo_2_code_patch_and_approval && ./run_demo.sh	Gemini 2.0 SQL patch → GitHub PR #142 → 1-Click Dashboard approval.
Demo 3: Precision Backfill	cd demos/demo_3_blast_radius_backfill && ./run_demo.sh	DAG lineage blast radius → BQ cost matrix (\$1.20) → Sequential backfill.
Demo 4: Feed Sentinel	cd demos/demo_4_source_feed_sentinel && ./run_demo.sh	GCS SLA arrival check → Auto-pause DAGs → File land detection & auto-resume.

Pre-Demo Checklist:

- [] All 13 GCP APIs enabled
- [] GCS Bucket gs://ctech-flowsentinel-demo-dev-landing-zone created
- [] BigQuery datasets raw_staging , dw_analytics , de_ops_metadata , backfill_sandbox created
- [] Dashboard accessible on http://localhost:3000 or Cloud Run URL